

Abdallah Abdelhady Khames

Data Engineer & ML Engineer

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SUMMARY

Third-year CS student (Data Science track, Alexandria University FCDS, expected 2027) who builds production-grade data platforms, ML pipelines, and BI systems measured in dollars and decisions, not accuracy scores. Work spans enterprise CRM/ERP analytics on medallion lakehouses, credit risk and fraud models, quantitative event studies, real-time computer vision, and a Power BI theme published to the Microsoft Fabric Community Gallery. Full case studies, live dashboards, and source code: abdallah-bodzz.github.io.

CORE SKILLS

Data Engineering: Medallion Architecture (Bronze/Silver/Gold), SCD Type 2, Star Schema, MDM, T-SQL, SQL Server, SQLAlchemy
ML & Quant: XGBoost, LightGBM, scikit-learn, lifelines, Monte Carlo / VaR / Expected Shortfall, OLS event studies, DCF & WACC
BI & Visualization: Power BI (DAX, drill-through, What-If, custom themes), Plotly, Chart.js, Excel (PivotTables, scenario modeling)
Backend & DevOps: Python (pandas, NumPy), FastAPI, Docker, Pydantic, Git, PyPI packaging
Computer Vision & NLP: OpenCV, MediaPipe, TensorFlow, LeIA (PT-BR NLP)
Databases: MS SQL Server, PostgreSQL, MySQL

FLAGSHIP PROJECTS

Enterprise CRM Intelligence & Customer 360 Platform | *SQL Server | Medallion Lakehouse | Python | Power BI* 2026

- Built a production Medallion Lakehouse on SQL Server with Customer MDM (SCD Type 2) across 96,096 customers, modeled after SAP CRM/Odoo CRM; five ML models feed a CRM Action Engine that prioritizes retention tasks.
- Delivered a 7-page Power BI report (29 DAX measures, R\$15.84M modeled GMV) and published its custom theme to the Microsoft Fabric Community Themes Gallery — 1,000+ views and downloads.

Enterprise Retail Analytics Platform | *SQL Server | Python | Power BI* 2026

- Built a 3-layer medallion lakehouse (7 staging → 8 warehouse → 10 mart) for a simulated ERP consulting engagement, with ABC classification (504 SKUs) and RFM/K-means segmentation (19,800 customers).
- Delivered 6 Power BI dashboards mapped to executive personas, with pre-aggregated mart metrics for sub-second load times.

Credit Risk Scoring & Fraud Detection API | *Python | XGBoost | LightGBM | FastAPI | Docker* 2024

- Loan-default model quantified at \$3.55M net benefit (152% ROI per 10,000 loans) via business-optimized threshold selection rather than accuracy.
- Fraud classifier (AUPRC 0.87 on 0.17% fraud-rate data) deployed as a FastAPI/Docker service, tuned to protect \$10,576 per run.

2026 Hormuz Blockade — Cross-Asset Shock Study (v2) | *Python | SciPy | yfinance | Plotly* 2026

- OLS event study of the 47-day Strait of Hormuz closure across 10 assets: WTI +32.9% while XOM fell -1.5% (beta flipped +0.15 → -0.42) — energy equities are not oil proxies under supply shocks.
- Quantified gold's -9.6% safe-haven failure; v2 adds a post-event recovery analysis on top of the original counterfactual baseline.

DriveGuard — Real-Time Driver Fatigue Detection | *Python | OpenCV | MediaPipe | TensorFlow* 2026

- Real-time drowsiness detection via webcam combining EAR/MAR geometry with a CNN hybrid (84k-image MRL dataset) — 98.8% accuracy, 3-5ms CPU inference, 25-30 FPS, fully local.

EXPERIENCE

IoT Software Development Trainee | *Alexandria University (FCDS) — Internship, On-site* Jul - Aug 2025

- Completed an intensive IoT Software Development Training program (Grade A): designed smart home systems integrating sensors, actuators, connectivity protocols, and cloud platforms.

Freelance Data Analyst / Processor | *Khamsat.com & Mostaqel — Remote* Jun - Dec 2025

- Completed 7 freelance data-cleaning and reporting contracts; added unrequested summary dashboards on two engagements and flagged 15+ pricing errors before a client publish — both acknowledged in client reviews.

Data Analytics Content Creator | *TikTok @data_w_bodzz | YouTube | Instagram* 2023 - Present

- 1,500+ followers and 200,000+ total views documenting Power BI builds, SQL walkthroughs, and project breakdowns; top video reached 16,000+ views.

EDUCATION

B.Sc. Computer Science, Data Science Specialization | *Alexandria University (FCDS), Egypt* Expected 2027

- DataCamp career tracks completed: Data Analysis in Python, SQL, Power BI, Machine Learning and Neural Networks.

LANGUAGES

Arabic: Native | English: Professional Working Proficiency